

1 Differential Forms

We recall some facts and propositions about differential forms.

For a differentiable manifold M of dimension n , $p \in M$, $(U; x^1, x^2, \dots, x^n)$ local coordinate of p . Then a differential k -form $\omega \in \Omega^k(M)$ can be locally expressed as

$$\omega = \sum_I a_I dx^I$$

where $a_I \in C^\infty(U)$, $I = (i_1, i_2, \dots, i_k)$ with $1 \leq i_1 < i_2 < \dots < i_k \leq n$ and $dx^I := dx^{i_1} \wedge dx^{i_2} \wedge \dots \wedge dx^{i_k}$. $d\omega \in \Omega^{k+1}(M)$ is hence locally given by

$$d\omega = \sum_I (da_I) \wedge dx^I = \sum_I \left(\sum_{i=1}^n \frac{\partial a_I}{\partial x^i} dx^i \right) \wedge dx^I.$$

We have some facts about the operator d :

- (1) $d \circ d = d^2 = 0$;
- (2) for $\omega \in \Omega^k(M)$ and $\tau \in \Omega^l(M)$, we have $d(\omega \wedge \tau) = (d\omega) \wedge \tau + (-1)^k \omega \wedge (d\tau)$.

Hence we get the homomorphism

$$d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$$

and the deRham complex

$$0 \longrightarrow \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \dots \xrightarrow{d} \Omega^n(M) \longrightarrow 0$$

We denote $\Omega^*(M) := \bigoplus_{k=0}^n \Omega^k(M)$. By the operation of wedge product

$$\wedge : \Omega^k(M) \times \Omega^l(M) \rightarrow \Omega^{k+l}(M)$$

$$(\omega, \tau) \mapsto \omega \wedge \tau$$

we can know that $\Omega^*(M)$ is a graded algebra.

Now suppose $F : N \rightarrow M$ is a smooth map between manifolds, $p \in N$, then we have the tangent map

$$\begin{aligned} F_* : T_p N &\rightarrow T_{F(p)} M \\ v &\mapsto F_*(v) \end{aligned}$$

where $F_*(v)(f) := v(f \circ F)$ for $\forall f \in C^\infty_{F(p)}(M)$. i.e., we can also write $F_* : TN \rightarrow TM$, $(p, v) \mapsto (F(p), F_*(v))$. And we also have the pullback map between the dual spaces of the tangent spaces

$$F^* : T_{F(p)}^* M \rightarrow T_p^* N$$

$$\omega_{F(p)} \mapsto F^*(\omega_{F(p)})$$

where $F^*(\omega_{F(p)})$ is given by $F^*(\omega_{F(p)})(v) := \omega_{F(p)}(F_*(v))$ for any $v \in T_p N$. Hence for $\omega \in \Omega^1(M)$, we define $(F^*\omega)_p := F^*(\omega_{F(p)})$ for any $p \in N$. (Recall that for a smooth function $g \in C^\infty(M) = \Omega^0(M)$, the pullback map of g by F is $F^*(g) := g \circ F$.)

Similarly, for $\omega \in \Omega^k(M)$, we define its pullback $F^*\omega$ as $(F^*\omega)_p := F^*(\omega_{F(p)})$, where $F^*(\omega_{F(p)})(v_1, v_2, \dots, v_k) = \omega_{F(p)}(F_*(v_1), F_*(v_2), \dots, F_*(v_k))$ for any $v_1, v_2, \dots, v_k \in T_p N$.

We have some properties about the pullback map:

Let $F : N \rightarrow M$ be a smooth map of manifolds. then

- (1) $\omega \in \Omega^k(M) \Rightarrow F^*\omega \in \Omega^k(N)$;
- (2) for $\omega, \tau \in \Omega^k(M)$ and $a \in \mathbb{R}$, we have $F^*(a\omega + \tau) = aF^*\omega + F^*\tau$;
- (3) $d \circ F^* = F^* \circ d$;
- (4) for $\omega, \tau \in \Omega^*(M)$, we have $F^*(\omega \wedge \tau) = F^*\omega \wedge F^*\tau$.

2 deRham Cohomology

We define the **deRham cohomology** as

$$H^q(M) := \frac{\ker\{d : \Omega^q(M) \rightarrow \Omega^{q+1}(M)\}}{\text{Im}\{d : \Omega^{q-1}(M) \rightarrow \Omega^q(M)\}}$$

For a smooth map $F : N \rightarrow M$, since $F^* \circ d = d \circ F^*$, we have the following commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \Omega^0(M) & \xrightarrow{d} & \Omega^1(M) & \xrightarrow{d} & \Omega^2(M) \xrightarrow{d} \dots \\ & & \downarrow F^* & & \downarrow F^* & & \downarrow F^* \\ 0 & \longrightarrow & \Omega^0(N) & \xrightarrow{d} & \Omega^1(N) & \xrightarrow{d} & \Omega^2(N) \xrightarrow{d} \dots \end{array}$$

note that F^* sends the closed forms to closed forms and exact forms to exact forms, thus we know that the pullback map F^* induces a homomorphism of the quotient spaces, also denoted by F^* :

$$F^* : H^q(M) \rightarrow H^q(N)$$

$$[\omega] \mapsto [F^*\omega]$$

where $\omega \in \ker\{d : \Omega^q(M) \rightarrow \Omega^{q+1}(M)\}$.

Recall the Zig-Zag Lemma, i.e, for a short exact sequence of cochain complexes

$$0 \longrightarrow \mathcal{A} \xrightarrow{i} \mathcal{B} \xrightarrow{j} \mathcal{C} \longrightarrow 0$$

which means that we have the following diagram with exact rows and columns as complexes:

$$\begin{array}{ccccccc}
 & & \vdots & & \vdots & & \vdots \\
 & & \uparrow d_{\mathcal{A}} & & \uparrow d_{\mathcal{B}} & & \uparrow d_{\mathcal{C}} \\
 0 & \longrightarrow & \mathcal{A}^{k+1} & \xrightarrow{i} & \mathcal{B}^{k+1} & \xrightarrow{j} & \mathcal{C}^{k+1} \longrightarrow 0 \\
 & & \uparrow d_{\mathcal{A}} & & \uparrow d_{\mathcal{B}} & & \uparrow d_{\mathcal{C}} \\
 0 & \longrightarrow & \mathcal{A}^k & \xrightarrow{i} & \mathcal{B}^k & \xrightarrow{j} & \mathcal{C}^k \longrightarrow 0 \\
 & & \uparrow d_{\mathcal{A}} & & \uparrow d_{\mathcal{B}} & & \uparrow d_{\mathcal{C}} \\
 0 & \longrightarrow & \mathcal{A}^{k-1} & \xrightarrow{i} & \mathcal{B}^{k-1} & \xrightarrow{j} & \mathcal{C}^{k-1} \longrightarrow 0 \\
 & & \uparrow & & \uparrow & & \uparrow \\
 & & \vdots & & \vdots & & \vdots
 \end{array}$$

then it gives rise to a long exact sequence in cohomology:

$$\dots \longrightarrow H^{q-1}(\mathcal{C}) \xrightarrow{d^*} H^q(\mathcal{A}) \xrightarrow{i^*} H^q(\mathcal{B}) \xrightarrow{j^*} H^q(\mathcal{C}) \xrightarrow{d^*} H^{q+1}(\mathcal{A}) \longrightarrow \dots$$

where the operator d^* is called the connecting homomorphism. By this lemma we obtain the following useful tool for computing the cohomology in manifolds:

Let $\{U, V\}$ be an open cover of the manifold M , then there are some inclusions

$$\begin{array}{ccc}
 & U & \\
 j_U \nearrow & & \searrow i_U \\
 U \cap V & & M \\
 j_V \searrow & & \nearrow i_V \\
 & V &
 \end{array}$$

and we have:

$$0 \longrightarrow \Omega^*(M) \longrightarrow \Omega^*(U) \oplus \Omega^*(V) \longrightarrow \Omega^*(U \cap V) \longrightarrow 0$$

$$\omega \longmapsto (\omega|_U, \omega|_V)$$

$$(\omega, \tau) \longmapsto \tau|_{U \cap V} - \omega|_{U \cap V}$$

which is an exact sequence and hence by zig-zag lemma, it gives rises to a long exact sequence of cohomology groups, called the **Mayer-Vietoris Sequence**:

$$\dots \longrightarrow H^{q-1}(U \cap V) \longrightarrow H^q(M) \longrightarrow H^q(U) \oplus H^q(V) \longrightarrow H^q(U \cap V) \longrightarrow H^{q+1}(M) \longrightarrow \dots$$

a useful tool for comupation in cohomology.

We have the following results about deRham cohomology.

(1) If the manifolds M and N are of the same homotopic type, then their deRham cohomology groups are isomorphic. (see An Introduction to Manifolds-Tu or Differential Forms in Algebraic Topology-Bott,Tu);

(2)**Theorem(deRham)**: Let M be a compact oriented smooth manifolds without boundary, then for any $0 \leq p \leq \dim M$, we have

a) $\dim H_{dR}^p(M; \mathbb{C}) < +\infty$;

b) $H_{dR}^q(M; \mathbb{C}) \cong H^q(M; \mathbb{C})$, where the latter denotes the singular cohomology.

(see Differential Forms in Algebraic Topology-Bott,Tu)

For a manifold M , the **Euler characteristic number** $\chi(M)$ is defined by

$$\chi(M) := \sum_{k=0}^{\dim M} (-1)^k \dim_{\mathbb{C}} H_{dR}^k(M; \mathbb{C}).$$

which is an important topological invariant of the manifolds.

3 Dolbeault Cohomology

Now, let M be a complex manifold of dimension n . Let $\pi : E \rightarrow M$ be a holomorphic vector bundle over M , define the vector bundle $T^{(p,q)}(E)$ as

$$T^{(p,q)}(E) := [\wedge^p T_{(1,0)}^*(M)] \wedge [\wedge^q T_{(0,1)}^*(M)] \otimes E,$$

where $T_{(1,0)}^*(M)$ is the dual bundle of the holomorphic tangent bundle $T_{(1,0)}(M) = \text{Span}_{\mathbb{C}}\{\frac{\partial}{\partial z^i}\}_{i=1}^n$ and $T_{(0,1)}^*(M)$ is the dual bundle of the antiholomorphic tangent bundle $T_{(0,1)}(M) = \text{Span}_{\mathbb{C}}\{\frac{\partial}{\partial \bar{z}^i}\}_{i=1}^n$ (Here, $(z^1, \dots, z^n)$ is a local coordinate of M and $\{\frac{\partial}{\partial z^i}\}_{i=1}^n, \{\frac{\partial}{\partial \bar{z}^i}\}_{i=1}^n$ form the basis of $T_{(1,0)}(M), T_{(0,1)}(M)$ respectively, thus we can know that $\{dz^i\}_{i=1}^n$ is the basis of $T_{(1,0)}^*(M)$ dual to $\{\frac{\partial}{\partial z^i}\}_{i=1}^n$ and similar for $\{d\bar{z}^i\}_{i=1}^n$).

From now on, for a holomorphic vector bundle E over the complex manifold M , we denote

$$A(U, E) := \{\text{smooth sections of } E \text{ on the open subset } U\}$$

$$\Gamma(U, E) := \{\text{holomorphic sections of } E \text{ on the open subset } U\}$$

Now let M be a complex manifold, E is a holomorphic vector bundle of rank r over M , $p \in M$, $\{e^1, \dots, e^r\}$ is a holomorphic local frame on an open nbhd U of p , $(z^1, \dots, z^n)$ is the local coordinate of U . Hence for $\forall \omega \in A(U, T^{(p,q)}(E))$ (i.e, ω is an E -valued (p, q) -form), ω may be written as

$$\omega = \sum_{\substack{i_1, \dots, i_p=1 \\ j_1, \dots, j_q=1}}^m \sum_{l=1}^r f_{i_1, \dots, i_p; j_1, \dots, j_q; l} dz^{i_1} \wedge \dots \wedge dz^{i_p} \wedge d\bar{z}^{j_1} \wedge \dots \wedge d\bar{z}^{j_q} \otimes e^l$$

If we define the operator $\bar{\partial}$ as

$$\begin{aligned}\bar{\partial} : A(U, T^{(p,q)}(E)) &\rightarrow A(U, T^{(p,q+1)}(E)) \\ w &\mapsto \bar{\partial}(w)\end{aligned}$$

where

$$\begin{aligned}\bar{\partial}(w) &:= \sum_{\substack{i_1, \dots, i_p=1 \\ j_1, \dots, j_q=1}}^m \sum_{l=1}^r \bar{\partial}(f_{i_1, \dots, i_p; j_1, \dots, j_q; l}) dz^{i_1} \wedge \dots \wedge dz^{i_p} \wedge d\bar{z}^{j_1} \wedge \dots \wedge d\bar{z}^{j_q} \otimes e^l \\ &= \sum_{\substack{i_1, \dots, i_p=1 \\ j, j_1, \dots, j_q=1}}^m \sum_{l=1}^r \frac{\partial(f_{i_1, \dots, i_p; j_1, \dots, j_q; l})}{\partial \bar{z}^j} d\bar{z}^j \wedge dz^{i_1} \wedge \dots \wedge dz^{i_p} \wedge d\bar{z}^{j_1} \wedge \dots \wedge d\bar{z}^{j_q} \otimes e^l\end{aligned}$$

it can be checked that such defined operator $\bar{\partial}$ is independent of the choice of the local holomorphic frame and the choice of the local coordinate, which means that the operator $\bar{\partial}$ is globally defined on the manifold M .

Now we denote $A^{(p,q)}(U, E) := A(U, T^{(p,q)}(E))$, then by the discussion above, we have obtained a linear operator

$$\bar{\partial} : A^{(p,q)}(U, E) \rightarrow A^{(p,q+1)}(U, E)$$

just like the occasion of deRham cohomology, we have the following discussions and results:

$$(1) \bar{\partial} \circ \bar{\partial} = 0$$

(2) Let E be a holomorphic vector bundle over a complex manifold M of dimension n , for any pair of non-negative integers (p, q) , we define

$$H^{(p,q)}(M, E) := \frac{\ker\{\bar{\partial} : A^{(p,q)}(M, E) \rightarrow A^{(p,q+1)}(M, E)\}}{\text{Im}\{\bar{\partial} : A^{(p,q-1)}(M, E) \rightarrow A^{(p,q)}(M, E)\}}$$

which is called the (p, q) -**Dolbeault Cohomology** of the holomorphic vector bundle E . (it's clear that $H^{(p,q)}(M, E) = 0$ if $p > m$ or $q > m$.) In particular, if $E = M \times \mathbb{C}$ is the trivial line bundle, then we just denote $A^{(p,q)}(M, M \times \mathbb{C}) =: A^{(p,q)}(M)$, $H^{(p,q)}(M, M \times \mathbb{C}) =: H^{(p,q)}(M)$ in which case the E -valued (p, q) -forms are just the ordinary (p, q) -forms on the manifold M . We denote $h^{(p,q)} := \dim H^{(p,q)} M$.

(3) if $F : M_1 \rightarrow M_2$ is a holomorphic map between complex manifolds and $\omega \in A^{(p,q)}(M_2)$, then $F^*(\omega) \in A^{(p,q)}(M_1)$ and we have $\bar{\partial} \circ F^* = F^* \circ \bar{\partial}$. i.e., for any $0 \leq p \leq \min\{\dim M_1, \dim M_2\}$, we have the commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & A^{(p,0)}(M_2) & \xrightarrow{\bar{\partial}} & A^{(p,1)}(M_2) & \xrightarrow{\bar{\partial}} & A^{(p,2)}(M_2) \xrightarrow{\bar{\partial}} \dots \\ & & \downarrow F^* & & \downarrow F^* & & \downarrow F^* \\ 0 & \longrightarrow & A^{(p,0)}(M_1) & \xrightarrow{\bar{\partial}} & A^{(p,1)}(M_1) & \xrightarrow{\bar{\partial}} & A^{(p,2)}(M_1) \xrightarrow{\bar{\partial}} \dots \end{array}$$

$$(4) \text{ for } \omega \in A^{(p_1, q_1)}(M) \text{ and } \tau \in A^{(p_2, q_2)}(M), \text{ we have } \bar{\partial}(\omega \wedge \tau) = \bar{\partial}(\omega) \wedge \tau + (-1)^{p_1+q_1} \omega \wedge \bar{\partial}(\tau).$$

(5) define the wedge product on Dolbeault cohomology as

$$\wedge : H^{(p_1, q_1)}(M) \times H^{(p_2, q_2)}(M) \rightarrow H^{(p_1+p_2, q_1+q_2)}(M)$$

$$([\omega, \tau]) \mapsto [\omega] \wedge [\tau] := [\omega \wedge \tau]$$

it can be checked that such defined operation is well-defined. Hence the linear space

$$\bigoplus_{r=0}^{2m} \bigoplus_{p+q=r} H^{(p, q)}(M)$$

becomes a graded algebra with respect to the wedge product defined above. By the similar discussion about that of deRham cohomology, such an algebra is biholomorphic invariant. i.e, if M_1 and M_2 are biholomorphic, then their algebraic structure of Dolbeault cohomology will be isomorphic.